



OPEN Innovative SVM optimization with differential gravitational fireworks for superior air pollution classification

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Amid escalating tension between environmental conservation and economic development, the imperative to enhance air quality has become increasingly urgent. This study elucidates a sophisticated approach for the assessment and remediation of air pollution issues through the integration of an enhanced particle swarm optimization algorithm and a differential gravitational fireworks algorithm-optimized support vector machine (SVM). In the initial phase of this research, a series of intricate data preprocessing and augmentation procedures were conducted, and the differential evolution algorithm played a pivotal role. The differential gravitational fireworks algorithm was subsequently introduced to optimize the SVM parameter settings, thereby bolstering classification accuracy and mitigating issues such as overfitting. Through rigorous and meticulous empirical testing, the augmented SVM model demonstrated notable performance in terms of classification accuracy and sequential and nonsequential data fusion, surpassing conventional SVM techniques. Notably, our sequential fusion method achieved an accuracy of up to 91%, at least 3% higher than that of nonsequential techniques. In conclusion, this study reveals an innovative and enhanced technological approach that is highly effective for the precise measurement and control of air pollution levels.

Keywords Differential evolution algorithm, Gravitational fireworks algorithm, Support vector machine (SVM), Sequential data fusion, Air quality classification, Optimization techniques

Economic reform in China has been a boon to the country's economic growth. However, the rough development model has contributed to severe air pollution. To address these problems, China has recently announced a series of air pollution control policies to strengthen air pollution control¹. Moreover, many cities are required to adopt air quality improvement policies² in the context of China's rapid economic development, increased industrial-scale expansion, and accelerated urbanization^{2,3}. Air pollution is a significant problem that affects people around the world. Accurate air quality analysis and evaluation can aid in controlling air pollution for improved urban planning and construction^{4,5}.

The air quality index (AQI, (2012)) adopted in this study divides air quality into six levels: excellent, good, lightly polluted, moderately polluted, heavily polluted, and severely polluted. In the calculation process, different pollutants are selected. However, the highest air quality subindex in the middle is the final air quality index. This method ignores the temporal and spatial characteristics of atmospheric pollutants. Air quality is affected by the concentration of multiple pollutants, and the AQI is biased in that it uses a single pollutant to assess air quality⁶.

In the wake of escalating global air pollution challenges, methods for air quality assessment have garnered substantial attention and application in recent years. The traditional methods, such as the analytic hierarchy process (AHP) and principal component analysis (PCA) employed by Zhang et al.⁷, yielded findings that were consistent with official environmental bulletins when applied to the atmospheric data of Xuzhou city. However, as data complexities have increased, researchers have begun delving into more sophisticated approaches.

Grey and cluster analyses have been recognized as potent tools in air quality assessments. For example, Ren⁸ introduced the novel ideal grey close function cluster correlation analysis, whereas Wang⁹ adopted an approach rooted in grey correlation clustering performance optimization. Moreover, other scholars have incorporated mathematical constructs into their evaluations; a notable example is Wang's¹⁰ integration of interval numbers for analysing the 2016 air quality of Liaoning Province.

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Conventional fuzzy evaluation techniques, exemplified by Wang's¹¹ two-grade index fuzzy comprehensive evaluation method, continue to find relevance in specific domains. However, the advent of machine learning and data-driven paradigms, as merged by Chang¹² with K-means clustering and Bayesian discriminant techniques, seem to overshadow these traditional avenues.

In light of the limitations inherent in the abovementioned strategies, novel methodologies harnessing the strengths of various approaches have emerged. Scholars such as Liang et al.¹³ and Yan et al.¹⁴ have ventured into artificial neural network methodologies. Additionally, considering the diverse and intricate nature of data, an array of methods spanning chromatographic analysis, PCA, and random forests have been proposed by researchers^{15–21}.

Air pollution data often exhibit nonlinear distribution characteristics due to the complex interactions between various pollutants and environmental factors. These nonlinearities pose significant challenges for traditional linear modelling techniques. To address these challenges, we employed a refined classification method that leverages the capabilities of support vector machines (SVMs), which are optimized via advanced metaheuristic algorithms. SVMs are particularly well suited for handling nonlinear and high-dimensional data because of their ability to map input features into higher-dimensional spaces through kernel functions, thereby facilitating the separation of complex data patterns. This attribute makes SVMs a strong choice for modelling the intricate relationships inherent in air pollution data.

Our study combines an SVM with the differential evolutionary strategy and the differential gravitational fireworks algorithm to enhance model performance. The differential evolutionary strategy helps explore the solution space effectively, whereas the differential gravitational fireworks algorithm optimizes the SVM parameters, thereby improving the classification accuracy and reducing the risks of overfitting and underfitting.

Furthermore, this study introduces an innovative sequential fusion classification technique. This method integrates multiple sources of data in a sequential manner, effectively capturing the continuity and intrinsic connections between data points. This approach addresses the limitations of traditional models by enhancing the understanding of data sequences and their interrelationships, leading to improved classification accuracy.

In addition to the core methodology, this manuscript introduces innovative techniques to enhance metaheuristic algorithms, specifically focusing on the integration of the differential evolutionary strategy with the differential gravitational fireworks algorithm. This enhancement aims to address common challenges in optimization problems, such as premature convergence and maintaining a balance between exploration and exploitation. By improving these algorithms, we provide more accurate and reliable classification predictions for air quality assessment.

Compared with the XGBoost method by Guo and Zheng²² and the deep learning approach by Zhou and Tan²³, our methodology demonstrates marked superiority in processing nonlinear and high-dimensional datasets. The crux of our contribution lies in providing a highly efficient and scientifically grounded solution to the existing challenges in air quality assessment. Our method not only increases computational performance but also enhances classification accuracy while effectively mitigating the risks of overfitting and underfitting.

Recent research findings in related fields also support the efficacy of our approach. For example, Wang et al.²⁴ demonstrated a novel approach to Alzheimer's disease prediction via a SVM, highlighting the versatility of SVMs in different applications. Houssein et al.²⁵ and Hu et al.²⁶ have made significant advancements in optimization techniques, which have direct implications for our work. Additionally, studies by Kurani et al.²⁷, Hou et al.²⁸, and Chen et al.²⁹ provide valuable insights into machine learning methods and optimization strategies that further substantiate our findings.

In contrast to these studies, our research introduces an enhanced SVM kernel that is optimized through a differential gravitational fireworks algorithm and is specifically tailored for air quality assessment. Liu et al.³⁰ utilized an SVM model for spatio-temporal air quality prediction, incorporating geographic features; however, they did not employ our differential gravitational fireworks optimization or sequential data fusion methods, which significantly enhance model robustness and prediction accuracy. Kouziokas³¹ developed a new SVM kernel combined with particle swarm optimization, focusing on PM10 prediction, whereas Li et al.³² established a predictive model for PM2.5 and NO₂ via quantum-behaved particle swarm optimization but did not address the sequential data fusion that we incorporate. Zapata-Hernandez et al.³³ proposed a method for predicting critical air quality events, achieving a 30% error rate in ozone predictions, indicating a need for greater sensitivity to pollutants. In contrast, in classifying both sequential and nonsequential data, our method demonstrated superior accuracy of up to 91%. Moreover, Murillo-Escobar et al.³⁴ successfully predicted multiple pollutants but did not leverage advanced optimization techniques to the extent that our research does. By integrating sophisticated optimization algorithms and increasing the SVM model's robustness, our approach addresses existing limitations in the literature, thus offering a significant improvement in air quality forecasting methodologies.

The synergistic integration of the differential evolutionary strategy with the differential gravitational fireworks algorithm offers precise optimization of the SVM parameters, which is pivotal in enhancing the accuracy and reliability of air quality evaluations.

Related theories

Differential evolution algorithm

In 1997, Rainer Storn and Kenneth Price proposed the differential evolutionary algorithm³⁵ on the basis of evolutionary ideas such as the genetic algorithm, a multiobjective (continuous variable) optimization algorithm. This algorithm is used to solve the global optimal solution in multidimensional space³⁶. Moreover, it offers several advantages, including simple principles, easy realization, strong global search ability, and fast convergence speed.

The differential algorithm includes four operation stages: initialization, mutation, crossover, and selection.

(1) Initialization

In the differential algorithm, the position of each individual represents a feasible solution.

$p = (p_1, p_2, \dots, p_d)$, taking N_P as the population number and d as the initial population number, and $i \in [1, N_P]$, $j \in [1, d]$ as the initial population number, the selection rules are as follows:

$$P_{i,j} = P_{\min,j} + rand_{i,j}(0, 1) * (P_{\max,j} - P_{\min,j}) \quad (1)$$

where $p_{\min,j}$ and $p_{\max,j}$ represent the upper and lower boundaries of the J dimension individual, respectively, and $rand_{i,j}(0, 1)$ is a generated random number between (0,1) of the J column in row I ³⁶.

(2) Mutation operation

The difference vector of two individuals was randomly selected from the population when starting from a randomly generated initial population, which serves as the source of the random variation of the third individual. This process occurs when the difference vector is weighted, and the third individual is added according to specific rules. The operation is called a mutation, and the formula is as follows:

$$V_i = X_{r1} + F * (X_{r2} - X_{r3}) \quad (2)$$

where r_1 and r_2 are randomly selected individuals in the population and F is the scaling factor³⁶.

(3) Cross factor

Crossover is the process of generating experimental individuals by mixing the parameters of the mutant with a predetermined target individual:

$$U_{ij} = \begin{cases} V_{ij} & \text{if } rand(0, 1) \leq CR \text{ or } j = j_{rand} \\ X_{ij} & \text{otherwise} \end{cases} \quad (3)$$

Each individual and its offspring are generated through a crossover. Specifically, for each component following a certain probability, the offspring variation vector (otherwise known as the original vector) is selected to generate test individuals³⁶.

(4) Selection

Suppose that the fitness value of the test individual is better than the fitness value of the target individual, the test individual will replace the target individual in the next generation. Otherwise, the target individual will be retained. This operation is called selection.

In the evolution process of each generation, each vector is used as the target individual. The algorithm keeps the excellent individual, eliminates the wrong individual, and guides the search process to approach the optimal global solution.

$$X_i^{G+1} = \begin{cases} U_i & \text{if } f(U_i) \leq f(X_i) \\ X_i & \text{otherwise} \end{cases} \quad (4)$$

The differential evolution algorithm adopts a greedy algorithm to select the better choice between the target individual and experimental individual according to the fitness function value³⁶.

Fireworks algorithm

Professor Tan Ying of Peking University proposed the fireworks algorithm (FWA) in 2010^{37–39}. According to Professor Tan, the FWA is a new swarm intelligence optimization algorithm inspired by fireworks exploding in the night sky to generate sparks and illuminate the surrounding natural phenomena. The position of each firework represents a feasible solution $X = (X_1, X_2, \dots, X_d)$. The firework algorithm includes three operations: explosion, mutation, and selection. Each firework explodes and produces several sparks, and in doing so, some will likely produce a unique Mars.

Sparks are randomly and evenly distributed around the current spark according to its amplitude. In contrast, mutated sparks are generated in a normal distribution near the present spark. The number of sparks generated in each iteration is greater than the number of each initial spark, and the process of firework explosion involves searching the neighbourhood. It calculates the amplitude and quantity of each firework through an interactive mechanism. The fireworks with good fitness values receive better resources, whereas those with poor fitness values receive fewer resources. The fireworks algorithm is characterized by locality, randomness, diversity, and complex implementation. Moreover, it falls into local search^{40,41}.

Gravitational search operator

Emit Rashedi and other scholars proposed the gravitational search algorithm (GSA) in 2009. This algorithm is a random heuristic search algorithm based on Newton's law of universal gravitation and the law of motion. Notably, the gravitational force between the particles solves the GSA, making the particles move towards the particle with an enormous mass. The particle with the largest mass occupies the optimal position, obtaining the

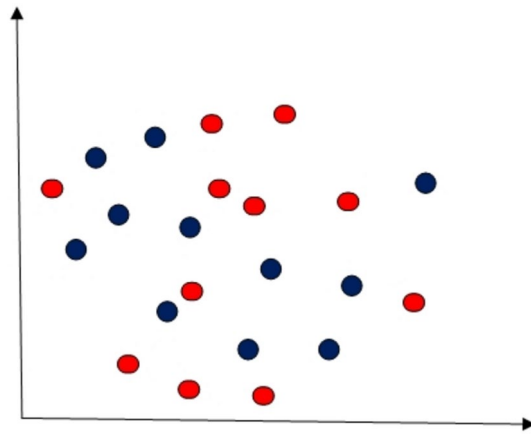


Fig. 1. Nonlinear support vector machine.

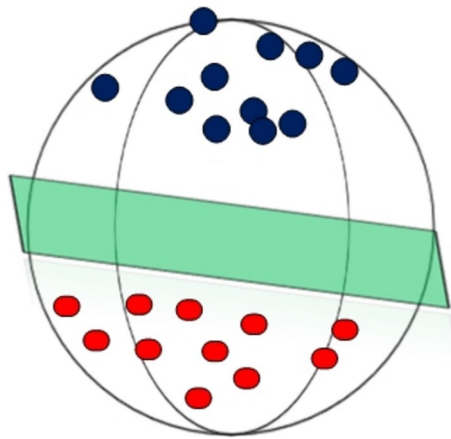


Fig. 2. Nonlinear support vector machine classification plane.

optimal solution to the optimization problem during movement. In this process, optimized information sharing is realized. The particles in the solution space can sense the global environment information, improve the poor latitude value, and guide the particle group to search for the optimal solution area.

Support vector machine

The SVM^{42–44} was first developed and proposed by Cortes and Vapnik in 1995. SVM is a new machine learning method based on statistical learning theory. It has unique advantages in solving small samples, nonlinear, and high-dimensional pattern recognition problems. Moreover, SVM can be extended to other machine learning problems, such as function fitting. The characteristic of the algorithm is to learn the sample data and obtain a maximum margin hyperplane of different class samples. In addition, the algorithm finds the best classification hyperplane among different class samples.

Many factors influence the spatial and temporal distributions and concentrations of air pollutants, resulting in nonlinear distributions⁴⁵. Guo et al.⁴⁶ suggested that vector machines can classify nonlinear data by introducing a kernel function and mapping the data from low-dimensional space (Fig. 1) to high-dimensional space (Fig. 2), followed by searching for a hyperplane in high-dimensional space.

Suppose the dataset in Fig. 1 is as follows:

$$T = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\} \quad (5)$$

where n is the number of samples, x_i is the input, and y_i is the output.

The SVM method conducts nonlinear mapping to the sample data and completes the transformation from low-dimensional space to high-dimensional space (Fig. 2). Afterwards, the SVM looks for the following hyperplanes in the high-dimensional space⁴⁷:

$$y = f(x) = w\phi(x) + b \quad (6)$$

where W is the weight vector, b is the bias component, and $\phi(x)$ represents a nonlinear mapping function that maps x to higher-dimensional pattern recognition.

We convert the hyperplane into a convex quadratic programming formula to find the optimal hyperplane solution:

$$\begin{cases} \min_{w,b,\xi_i} (\frac{1}{2} \|w\|^2 + c \sum_{i=1}^n \xi_i) \\ y_i(w * \phi(x_i) + b) \geq 1 - \xi_i, \\ \xi_i \geq 0, (i = 1, 2, \dots, n) \\ \forall i = 1, \dots, n \end{cases} \quad (7)$$

where c is the penalty factor, indicating the tolerance of abnormal values, and ξ_i is a relaxation variable, a high-dimensional space kernel function $K(x_i, x_j) = \phi(x_i)\phi(x_j)$.

Lagrangian multipliers are introduced to solve (5) better, and their expression becomes the following:

$$L(w, b, a, \xi) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n a_i [y_i(w * \phi(x_i) + b) - 1 + \xi_i] - \sum_{i=1}^n u_i \xi_i \quad (8)$$

The selection of kernel functions is as follows:

$$k(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (9)$$

The expression of the discriminant function is as follows:

$$f(x) = \text{sgn}(\sum_{i=1}^m a_i y_i K(x_i, x) + b) \quad (10)$$

where g is the kernel representing the width of the radial basis function action and c is the penalty factor. The loss function ε , penalty parameter c , and kernel function parameter g affect the performance of the SVM. If the penalty parameter c is too small, it will lead to underlearning. If it is too large, it will lead to overlearning. Thus, g is a kernel function parameter that reflects the characteristics of the training set and determines the understanding complexity. Therefore, the selection of parameters plays a vital role in classification efficiency and accuracy. This research uses differentials to improve the population, the SVM gravitation fireworks optimization algorithm, and parameter optimization.

Sequential fusion classification

In this study, we also introduce an innovative sequential fusion classification technique. This method integrates multiple sources of data in a sequential manner, effectively capturing the continuity and intrinsic connections between data points. This approach addresses the limitations of traditional models by enhancing the understanding of data sequences and their interrelationships, leading to improved classification accuracy.

Improved algorithm: Differential gravitational fireworks algorithm

Some individual problems, such as fireworks and sparks, are not well informed when searching for an optimum solution. The algorithm yields a poor result when applied to shifted functions whose optimum is not at or near the origin. Thus, a hybrid fireworks algorithm with a gravitational search operator⁴⁸ is proposed. Through gravitation, the particles interact to produce new sparks, which increases the diversity of sparks, improves the algorithm search accuracy and convergence speed, and reduces the possibility of falling into a local search.

This study uses the gravitational search operator to optimize the fireworks algorithm. The specific steps of the gravitation fireworks algorithm are as follows:

- (1) Parameters such as the number of iterations and fireworks are set and each firework position is initialized³⁷.
- (2) The amplitude of the explosion for each firework A_i is calculated. The calculation is shown in (11):

$$A_i = \frac{S * (f(x_i) - f_{\min})}{\sum_{j=1}^n (f(x_j) - f_{\min}) + \varepsilon} \quad (11)$$

where A_i is the amplitude, S is a constant, $f(x_i)$ is the fitness value of the firework, $f_{\min}$ is the minimum fitness value, and ε is a small constant to avoid division by zero⁴⁹.

- (3) The number of sparks that each firework can generate is determined as follows:

$$N_i = m * (1 - \frac{f(x_i) - f_{\min}}{\sum_{j=1}^n (f(x_j) - f_{\min}) + \varepsilon}) \quad (12)$$

where m is a constant for limiting the total number of sparks generated³⁷.

(4) The spark particles out of the range are mapped into the boundary range according to the following:

$$x'_k = x_k + \text{rand}(0, 1) * (x_{\max,k} - x_{\min,k}) \quad (13)$$

where x_k is the spark particle, and $x_{(\max,k)}$ and $x_{(\min,k)}$ are the upper and lower bounds of the solution space in dimension K , respectively, revealing that this is a modular operation.

(5) The gravitational force between particles is calculated:

$$F_{ij} = \frac{G * M_i * M_j}{R_{ij} + \varepsilon} \quad (14)$$

where F_{ij} is the force, G is the gravitational constant, M_i and M_j are the masses of particles i and j , R_{ij} is the Euclidean distance between the particles, and ε is a small constant⁴⁶.

(6) The new position of the particle is updated according to the following:

$$X_i^{\text{new}} = X_i + \text{rand}(0, 1) * F_i \quad (15)$$

(7) If the number of iterations is reached, the iteration is stopped, and the optimal result is output. Otherwise, the iteration is continued.

Fitness calculation

The fitness function for the optimization algorithm is crucial for evaluating the quality of each solution. For our study, the fitness function $f(x)$ is defined as the inverse of the mean squared error (MSE) of the SVM predictions. This ensures that solutions with lower prediction errors are preferred. MSE is calculated as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (16)$$

where y_i is the actual value and $\hat{y}_i$ is the value predicted from the SVM model. 2.7 SVM performance assessment and hyperparameter optimization.

To assess the performance of SVM and optimize its hyperparameters, we employ a grid search combined with cross-validation. The key hyperparameters for SVM include the penalty parameter C and the kernel parameter γ . The steps for optimization are as follows:

- (1) Grid search: A range of values for C and γ are defined. For example, $C \in [0.1, 1, 10, 100]$ and $\gamma \in [0.001, 0.01, 0.1, 1]$.
- (2) Cross-validation: For each combination of C and γ , k -fold cross-validation is performed (typically $k = 5$ or $k = 10$). The data is split into k subsets, SVM is trained on $K - 1$ subsets and validated on the remaining subset. This process is repeated k times, and the average performance metric (e.g., accuracy or F_1 score) is calculated.
- (3) Selection: The combination of C and γ that yields the best cross-validation performance is chosen. This ensures that the model is optimized for generalization to new, unseen data.
- (4) Final model training: The optimal C and γ is used to train SVM on the entire training dataset. The final model is evaluated on a separate test set to ensure its performance.

Experimental settings and environment

To ensure the reliability of the experimental results, we conducted each experiment independently a minimum of 10 times. This approach helps mitigate the influence of randomness and ensures that the results are robust and consistent. The experimental settings for the differential gravitational fireworks algorithm included a population size of 50 and 100 iterations. The experiments were conducted on a computer running Windows 10 and equipped with an Intel Core i7-9700 K processor and 32 GB of RAM. The software environment consisted of Python 3.8 and the Scikit-learn library for SVM implementation.

Differential gravitational fireworks algorithm

This section introduces the improved differential gravitational fireworks optimization (DGFO) algorithm, which combines the strengths of differential evolution, gravitational search, and fireworks algorithms to optimize SVM parameters. The pseudocode of the DGFO algorithm is provided below to illustrate the steps involved in the optimization process.

Level pollutant	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7
PM _{2.5}	35	75	115	150	250	350	500
PM ₁₀	50	150	250	350	420	500	600
SO ₂	50	150	475	800	1600	2100	2620
NO ₂	40	80	180	280	565	750	940
CO	2	4	14	24	36	48	60
O ₃ ₈	100	160	215	265	800	1000	1200

Table 1. Air pollution concentration standards.

AQI values	AQI level	AQI level	Level color	Health impact
0–50	1	Excellent	Green	The air quality is very good/comfortable
51–100	1	Good	Yellow	The air quality is good, but for a very small number of sensitive groups, individual pollutants may have health effects
101–150	2	Light pollution	Orange	Sensitive people's symptoms will become worse, and healthy people may also have symptoms of heart and respiratory discomfort
151–200	3	Moderate pollution	Red	Sensitive people will feel serious discomfort, and healthy people may also have symptoms of heart and respiratory discomfort
201–300	4	Heavy pollution	Purple	The symptoms of patients with heart disease may be aggravated; very unsuitable for exercise
> 300	5	Severe pollution	Maroon	Not suitable for exercise for healthy people; particularly obvious and strong symptoms may occur, as well as some diseases

Table 2. Air quality standards.

Differential Gravitational Fireworks Algorithm Pseudocode.
Initialize population P with random solutions.
Set parameters: number of iterations, S, m, G, epsilon.
While termination criteria not met do:
 Evaluate fitness of each firework in P³⁶.
 Calculate explosion amplitude A and number of sparks N for each firework⁴⁹.
 Generate sparks and map out-of-range sparks into boundary range³⁷.
 Calculate gravitational force F between particles⁵⁰.
 Update positions of fireworks based on F³⁷.
 Select the best fireworks for the next generation⁴⁹.
End While.
Return the best solution f.

Methods
Pollutant concentration and classification standard

The evaluation set {I, II, III, IV, V, VI, VII} follows the standard air pollution concentration stipulated by the State Environmental Protection Bureau. The higher the grade is, the higher the pollution concentration and the more significant the impact on health. The levels of different pollutants are listed in Table 1, and the AQI levels are listed in Table 2:

Data preprocessing

We collected data from the Qingyue Open Environment Data Centre. We also collected real-time monitoring data of air pollution in Xi'an in 2022 as the initial concentration values for air pollutants. The mass concentration units of SO₂, NO₂, O₃, PM₁₀, and PM_{2.5} are µg/m3, and the mass concentration units of CO are mg/m3. Table 3 shows 26 monitoring areas:

Monitoring values of some air pollution will be missing during monitoring because of the failure of atmospheric pollution monitoring acquisition equipment or environmental factors, and these missing values will affect decision-making regarding air pollution control. Thus, we use the average method to fill in the missing values and reduce the errors in air pollution decision-making, as shown in Table 4.

Classification of air pollution concentration data using the differentially improved population and SVM gravitational fireworks optimization

We classify air pollution concentration data using the differentially improved population and SVM gravitational fireworks optimization algorithm. The details are as follows:

The model is divided into four modules: the preprocessing data module, population size expansion, parameter optimization module, and SVM module. The data preprocessing module divides and normalizes the data. Then, the model expands the population size via the differential evolution algorithm to generate a new population, expands the population size, and optimizes the parameters. The model uses the gravitational fireworks algorithm

SN	NO ₂	CO	O ₃	PM ₁₀	PM _{2.5}	Quality	Pubtime
1	55	1.756	16	172	102	III	2021-1-3 0:00
2	104	1.411	3	185	130	IV	2021-1-3 0:00
3	106	2.072	4	217	147	IV	2021-1-3 0:00
4	91	1.99	3	268	165	V	2021-1-3 0:00
5	88	1.838	8	253	203	V	2021-1-3 0:00
6	94	1.366	4	158	115	III	2021-1-3 0:00
7	86	1.068	4	183	131	IV	2021-1-3 0:00
8	83	1.656	5	197	141	IV	2021-1-3 0:00
9	129	3.339	4	291	115	IV	2021-1-3 0:00
10	87	1.976	5	150	119	IV	2021-1-3 0:00
11	91	1.65	3	354	179	V	2021-1-3 0:00
12	86	1.937	23	166	105	III	2021-1-3 0:00
13	99	1.661	2	210	162	V	2021-1-3 0:00
14	78	0.904	6	187	76	III	2021-1-3 0:00
-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-
196	78	1.547	3	298	186	V	2021-1-3 7:00
197	47	0.776	16	131	50	II	2021-1-3 7:00
198	96	1.675	6	358	167	V	2021-1-3 7:00
199	52	1.495	2	140	93	III	2021-1-3 7:00
200	85	1.399	8	148	126	IV	2021-1-3 7:00
201	74	1.595	2	232	187	V	2021-1-3 7:00
202	41	1.025	36	89	84	III	2021-1-3 7:00
203	76	1.252	2	159	86	III	2021-1-3 7:00
204	74	1.685	2	141	94	III	2021-1-3 7:00
205	78	1.289	5	191	128	IV	2021-1-3 7:00
206	73	1.599	3	182	116	IV	2021-1-3 7:00
207	95	1.771	3	158	115	III	2021-1-3 7:00

Table 3. Monitored air pollution concentrations.

to obtain the optimal combination of SVM penalty parameters and kernel parameters. The steps are shown in Fig. 3.

The steps are as follows:

- (1) The training and test sets are divided, and the input data is normalized.
- (2) The population size, scaling factor, cross probability, boundary value, and other parameters are initiated.
- (3) The parameters of the SVM kernel function and penalty factor are set to optimize the range and calculation precision. The initial optimal objective and optimal solution are determined.
- (4) The mutation operator is generated iteratively using the differential algorithm.
- (5) If the mutation operator exceeds the boundary value, it is the boundary value. If it is less than the boundary value, it is the newly generated value, forming 2n populations.
- (6) The fireworks population 2n is initialized to find the corresponding targets of all fireworks and spark cycles. These targets are compared with the maximum and minimum to obtain the final maximum and minimum and limit the boundary value.
- (7) The inertial mass of all fireworks and sparks is calculated.
- (8) The 2n fireworks with the largest inertial mass are selected to calculate their mutual gravitation. Thus, the fireworks are moved according to gravitation.
- (9) The fitness is calculated, and the optimal solution is updated.
- (10) The current iteration number is judged on whether it reaches the termination condition of the set algorithm. If yes, the combination of the best error penalty parameter c and the kernel function parameter g searched by the gravitation fireworks algorithm is output. If not, the algorithm returns to Step 4 to continue iterating.
- (11) The best penalty parameter c and kernel parameter g are obtained.
- (12) The SVM training model is constructed using the optimal (c,g).
- (13) The SVM training model is applied to classify air pollution concentrations.
- (14) The output classification results are compared with the actual classification results, and the classification accuracies are compared.

SN	NO ₂	CO	O ₃	PM ₁₀	PM _{2.5}	Quality	Pubtime
1	55	1.756	16	102	172	3	2021-1-3 0:00
2	104	1.411	3	130	185	4	2021-1-3 0:00
3	106	2.072	4	147	217	4	2021-1-3 0:00
4	91	1.99	3	165	268	5	2021-1-3 0:00
5	88	1.838	8	203	253	5	2021-1-3 0:00
6	94	1.366	4	115	158	3	2021-1-3 0:00
7	86	1.068	4	131	183	4	2021-1-3 0:00
8	83	1.656	5	141	197	4	2021-1-3 0:00
9	129	3.339	4	115	291	4	2021-1-3 0:00
10	87	1.976	5	119	150	4	2021-1-3 0:00
11	91	1.65	3	179	354	5	2021-1-3 0:00
12	86	1.937	23	105	166	3	2021-1-3 0:00
13	99	1.661	2	162	210	5	2021-1-3 0:00
14	78	0.904	6	76	187	3	2021-1-3 0:00
-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-
196	78	1.547	3	186	298	5	2021-1-3 7:00
197	47	0.776	16	50	131	2	2021-1-3 7:00
198	96	1.675	6	167	358	5	2021-1-3 7:00
199	52	1.495	2	93	140	3	2021-1-3 7:00
200	85	1.399	8	126	148	4	2021-1-3 7:00
201	74	1.595	2	187	232	5	2021-1-3 7:00
202	41	1.025	36	84	89	3	2021-1-3 7:00
203	76	1.252	2	86	159	3	2021-1-3 7:00
204	74	1.685	2	94	141	3	2021-1-3 7:00
205	78	1.289	5	128	191	4	2021-1-3 7:00
206	73	1.599	3	116	182	4	2021-1-3 7:00
207	95	1.771	3	115	158	3	2021-1-3 7:00

Table 4. Monitored air pollution concentrations.

Model analysis

Our data comprise air pollution concentration data, which are nonlinearly distributed. When SVM is used for classification, the introduction of a kernel function is vital for data classification accuracy and for mapping data from low to high dimensions. The effect of the SVM classification algorithm is closely related to the selection of the kernel function, kernel parameters, and penalty coefficient. Thus, the greater the penalty coefficient is, the smaller the tolerance for error. The greater the tolerance for error is, the smaller the penalty coefficient and fitting degree. The larger the kernel parameter is, the weaker the segmentation ability of the kernel function. Therefore, the selection of the penalty coefficient and kernel parameters affects the classification and generalization ability of the SVM algorithm. This study proposes the differential gravitational fireworks algorithm to optimize SVM support based on the following considerations.

- (1) The pyrotechnic algorithm with gravitation generates new sparks through interaction with the dominant particles, which increases spark diversity and improves the search accuracy and convergence speed of the algorithm. This process makes the algorithm have strong local search ability.
- (2) The differential algorithm based on the genetic algorithm is used to solve the optimal global solution in high-dimensional feature space. This algorithm is characterized by fast convergence, no premature convergence, strong robustness, and substantial global search ability^{13,14,21}.
- (3) Obvious advantages are derived by combining the differential algorithm with the gravitational algorithm to optimize the fireworks algorithm. This method expands the population size through the global optimization ability of the differential evolution algorithm and the development ability of the gravitational search algorithm population in the later stage of evolution. The fireworks algorithm can be optimized in a larger area, as it improves the global optimization ability of the fireworks algorithm. The algorithm uses the convergence accuracy, convergence speed, and strong local search ability of the gravitational fireworks algorithm to quickly find the optimal combination of penalty and kernel parameters.

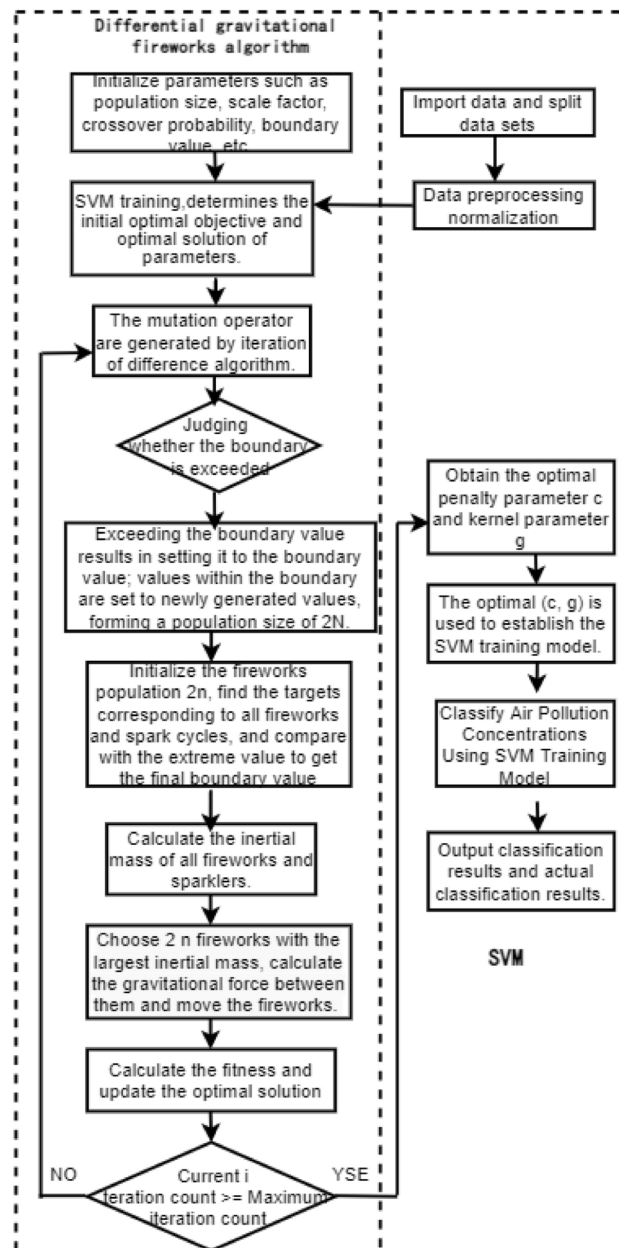


Fig. 3. Differential gravitational fireworks SVM.

Simulation

Realization of SVM differential gravitational fireworks optimization via MATLAB

In the present research, we utilized MATLAB to develop and execute a classification algorithm for support vector machines (SVMs), which was optimized through a novel differential gravitational fireworks method. This algorithm was employed to analyse the dataset outlined in Table 3, which was judiciously partitioned into training and testing subsets using Kfold cross-validation for rigorous model validation.

The core difference between sequential and nonsequential fusion classification lies in the treatment of data interdependencies. Sequential fusion classification, adapted from the second law of geography, emphasizes the interdependencies among datapoints in a sequence, treating them as vital elements during the classification process. This sequential nature allows the model to recognize patterns and relationships over consecutive data points, potentially increasing classification accuracy. In contrast, nonsequential fusion classification disregards such sequential correlations, often leading to oversimplifications and potential misclassifications.

To demonstrate the advantages of sequential fusion, we conducted a comparative analysis of three distinct classification techniques: the SVM optimized by differential gravitational fireworks, as presented in this study, the SVM optimized by gravitational fireworks, and the traditional SVM. The goal was to highlight the superiority of the SVM optimized by differential gravitational fireworks in addressing similar problems and to accentuate the significant increase in precision of the sequential fusion classification technique compared with

its nonsequential counterpart. This comparison reveals that the sequential method consistently outperforms the nonsequential approach, providing both theoretical and practical justifications for researchers in related fields.

Sequential and nonsequential classification simulation based on a support vector machine

Sequential classification method Using MATLAB, we established a support vector machine method that implemented a sequential fusion methodology for predicting the air quality index. The simulation outcomes, represented in Figs. 4–7, successively applied atmospheric pollution concentration data from the previous period for model training and sequentially predicted the atmospheric pollution concentration values for the subsequent period. The performance of the model was evaluated by measuring the discrepancy (e.g., mean squared error) between the actual and predicted values.

Nonsequential classification method Using MATLAB as a foundation, we developed a support vector machine method that employed a nonsequential fusion technique for predicting the air quality index. As shown in Fig. 8, we trained the model using atmospheric pollution concentration values from the previous period and tested it on values from the following period. The performance of the model was evaluated by measuring the discrepancy (e.g., mean squared error) between the actual and predicted values.

Sequential and nonsequential classification simulation based on a gravitational fireworks-optimized support vector machine

Sequential classification method We built a gravitational fireworks support vector machine (SVM) method using MATLAB, employing sequential fusion to predict the air quality index. As illustrated in Figs. 9–12, atmospheric pollution concentration data from the previous period were applied in the training of the model, and the sequential prediction of atmospheric pollution concentration values for the subsequent period was conducted.

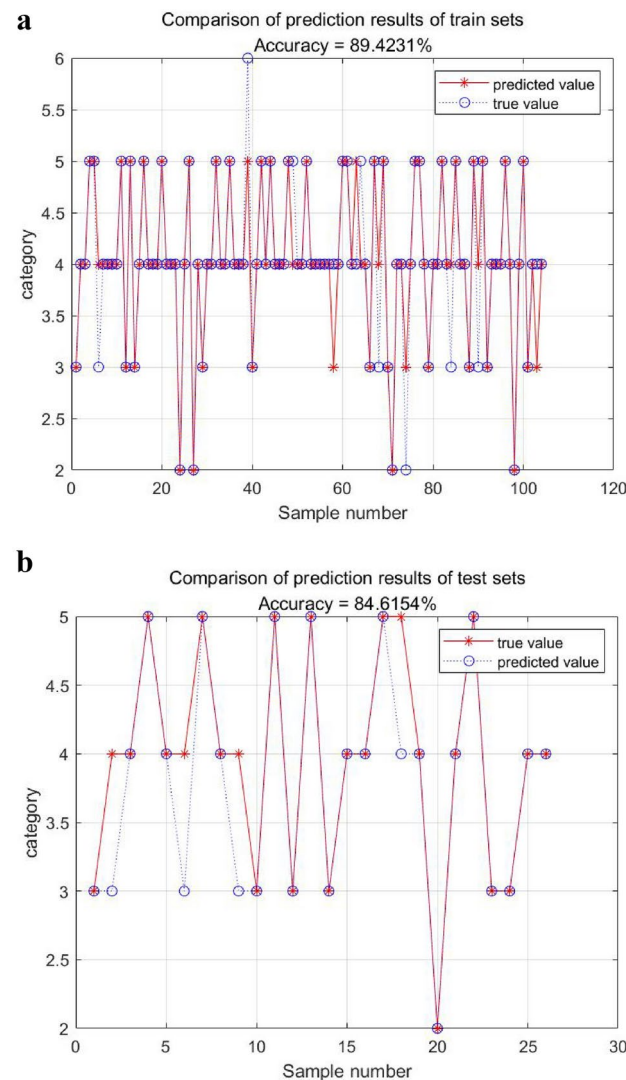


Fig. 4. (a) Comparison of the 4-point results based on the support vector machine classification training set. (b) Comparison of the 4-point results based on the support vector machine classification test set.

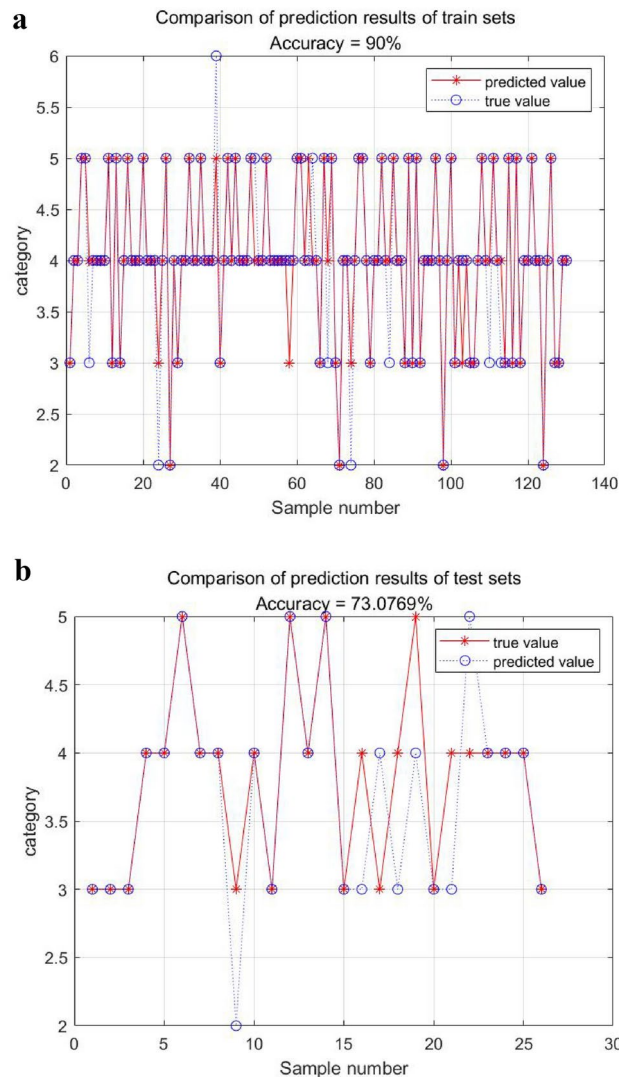


Fig. 5. (a) Comparison of the 5-point results based on the support vector machine classification training set. (b) Comparison of the 5-point results based on the support vector machine classification test set.

The performance of the model was evaluated by calculating the discrepancy (e.g., mean squared error) between the actual and predicted values.

Nonsequential classification A gravitational fireworks SVM method was constructed using MATLAB, adopting a nonsequential fusion technique for predicting the air quality index. As depicted in Fig. 13, atmospheric pollution concentration values from the previous period were utilized to train the model, which was then tested on the atmospheric pollution concentration values of the subsequent period. The performance of the model was evaluated by calculating the discrepancy (e.g., mean squared error) between the actual and predicted values.

Sequential and nonsequential classification simulation of the optimal support vector machine based on differential gravitational fireworks

Sequential classification method Using MATLAB, we developed a differential gravitational fireworks SVM method and applied sequential fusion to predict the air quality index. As illustrated in Figs. 14–17, we utilized atmospheric pollution concentration data from the prior period to train the model and then made sequential predictions for the atmospheric pollution concentration values of the subsequent period. The performance of the model was evaluated by calculating the discrepancy (e.g., mean squared error) between the actual and predicted values.

Nonsequential classification method A differential gravitational fireworks SVM method was developed using MATLAB, which incorporated a nonsequential fusion approach for predicting the air quality index. As depicted in Fig. 18, atmospheric pollution concentration values from the previous period were used to train the model,

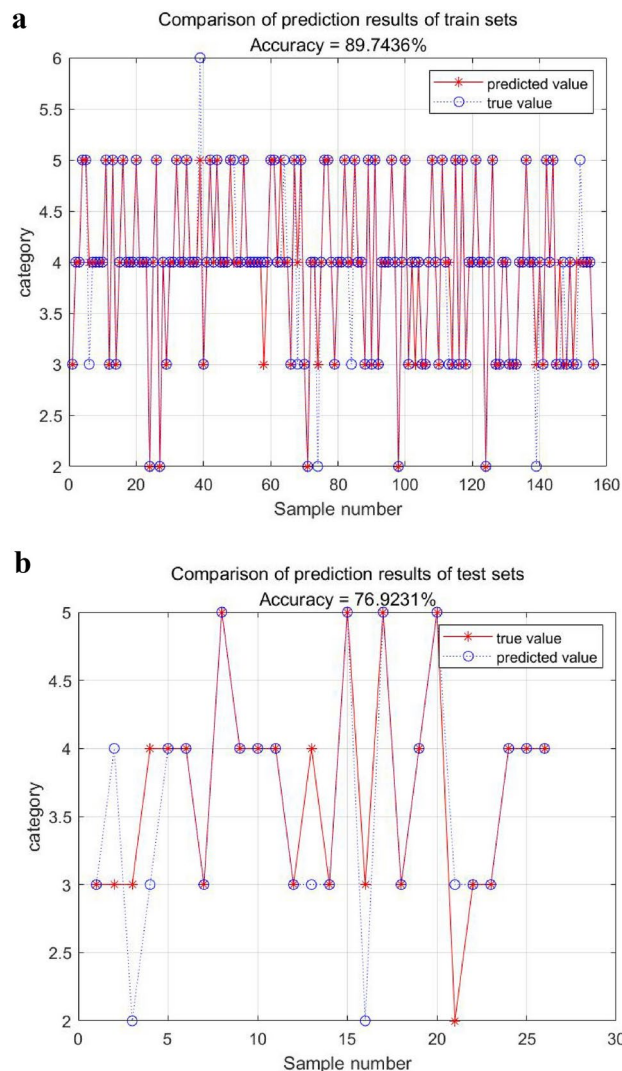


Fig. 6. (a) Comparison of the 6-point results based on the support vector machine classification training set. (b) Comparison of the 6-point results based on the support vector machine classification test set.

and then values for the subsequent period were obtained. The performance of the model was evaluated by calculating the discrepancy (e.g., mean squared error) between the actual and predicted values.

Comparative conclusion

Through in-depth simulation experiments, we extracted several pivotal insights. As shown in Table 5, first, the sequential fusion classification method significantly outperforms nonsequential fusion classification in terms of performance. This augmentation, which is primarily quantified via benchmarks such as the mean squared error, can be attributed to the innovative capacity of the sequential method to comprehend and harness the continuity of data sequences and the intrinsic interconnections between data points—qualities not effectively addressed by traditional models.

In juxtaposition with prevalent techniques such as deep neural networks (DNNs) and recurrent neural networks (RNNs), our method underscores its versatility. While DNNs and RNNs are lauded for their profound representational power, they often necessitate substantial computational resources and extensive training datasets. In contrast, our sequential fusion classification approach strikes a harmonious balance between performance and computational demands. Additionally, the adaptability of our method is particularly advantageous in scenarios where dataset size and diversity may be constrained, positioning it as a formidable contender in the realm of classification models.

Furthermore, the support vector machine (SVM) approach we have introduced, which is fortified by differential gravitational fireworks optimization, shows distinct advantages over existing SVMs integrated with gravitational fireworks optimization and conventional SVMs. To underscore the efficacy of our SVM model, which is supercharged with differential gravitational fireworks optimization, a classification accuracy of 91% was achieved, eclipsing other mentioned methodologies. The superiority of our method is reflected through

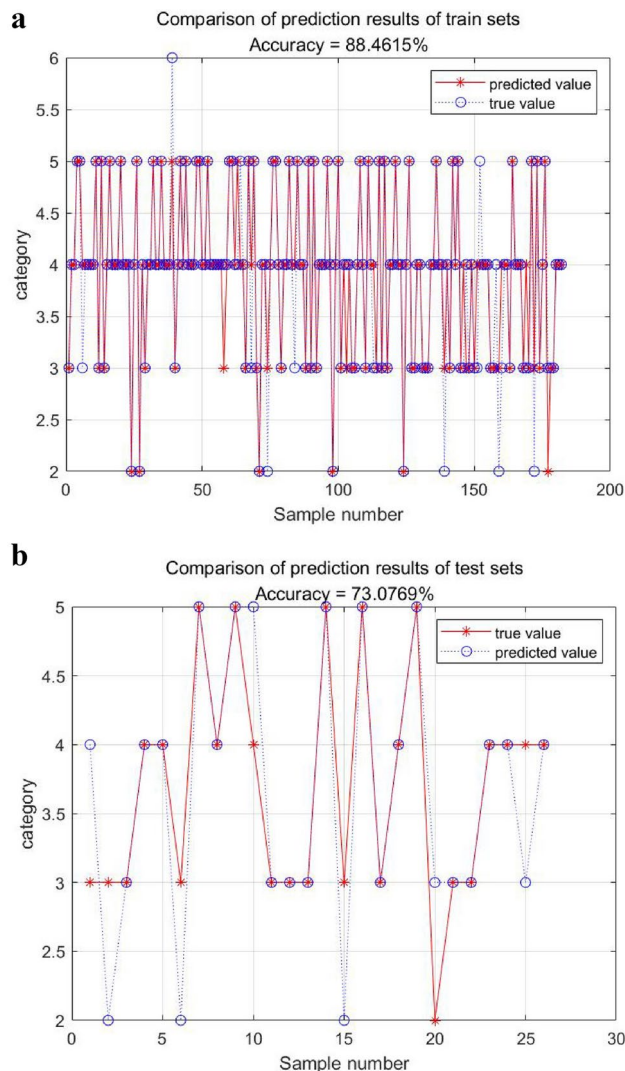


Fig. 7. (a) Comparison of the 7-point results based on the support vector machine classification training set. (b) Comparison of the 6-point results based on the support vector machine classification test set.

these metrics and highlights its unparalleled proficiency in addressing complex classification challenges, further accentuating the unique merits of sequential fusion classification in increasing classification accuracy.

In conclusion, our empirical findings strongly support the advantages of the sequential fusion classification technique. Particularly in scenarios demanding intricate data patterns and rigorous classification stipulations, our methodology clearly stands out, remedying the shortcomings that are inherent to traditional models. Thus, this work not only enriches current academic discourse but also provides a clear trajectory for future research pursuits.

Experimental setup and results

Experimental setup

To comprehensively evaluate the performance of our proposed differential gravitational fireworks optimized support vector machine (*DGFO-SVM*) method, we conducted experiments on a benchmark air quality dataset. This dataset included various pollutant concentration data collected from multiple monitoring stations. The goal was to classify air quality into predefined categories based on these measurements.

The experimental settings included the air quality dataset as the primary data source, which consisted of pollutant concentrations, and the target variable was the air quality index (*AQI*) category. The evaluation metrics used to assess the performance of the model were accuracy, F_1 score, precision, and recall.

To demonstrate the effectiveness of our differential gravitational fireworks optimized support vector machine (*DGFO-SVM*) method, we compared it with the following classification algorithms: a baseline support vector machine (*SVM*), a gravitational fireworks optimized SVM (*GFO-SVM*), a fully connected deep neural network (*DNN*), a recurrent neural network (*RNN*) with LSTM cells, XGBoost (an implementation of gradient boosted decision trees), and a random forest (*RF*) ensemble learning method based on decision trees.

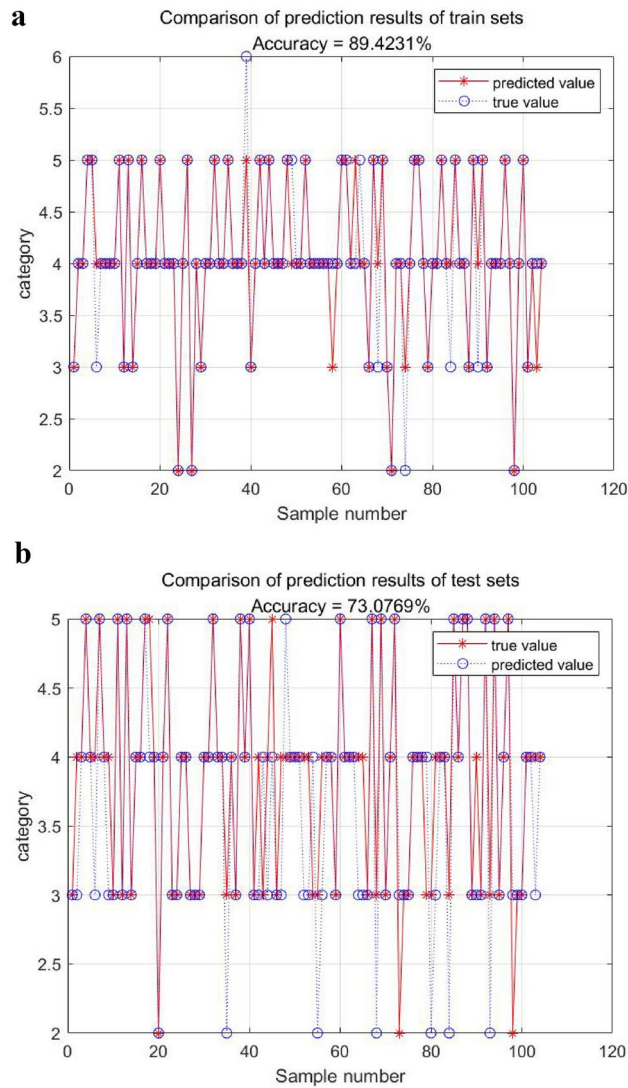


Fig. 8. (a) Comparison of the 4–7-point results based on a nonsequential classification training set of support vector machines. (b) Comparison of the 4–7-point results based on a nonsequential classification test set of support vector machines.

The experiments were conducted on a benchmark air quality dataset that included various pollutant concentration data collected from multiple monitoring stations. The goal was to classify air quality into predefined categories based on these measurements. The features consisted of pollutant concentrations, and the target variable was the air quality index (*AQI*) category. The evaluation metrics used to assess the performance of the model were accuracy, F_1 score, precision, and recall.

The experimental environment was set up on a computer running Windows 10 and equipped with an Intel Core i7-9700 K processor and 32 GB of RAM. The software environment consisted of Python 3.8, TensorFlow 2.4, and Scikit-learn 0.24.

Results and analysis

We evaluated the performance of each algorithm using a tenfold cross-validation approach to ensure the robustness and reliability of the results. Tables 6, 7 present a summary of the performance metrics for each algorithm.

Analysis

The results clearly demonstrate that our proposed *DGFO – SVM* method outperforms the baseline *SVM* and other comparison algorithms in terms of accuracy and F_1 score. Notably, *DGFO – SVM* achieves an accuracy of 91%, which is at least 2% higher than that of the best-performing alternative (*RNN*) and significantly better than that of the traditional alternative *SVM*. These results validate the effectiveness of integrating the differential gravitational fireworks optimization with *SVM* for air quality classification.

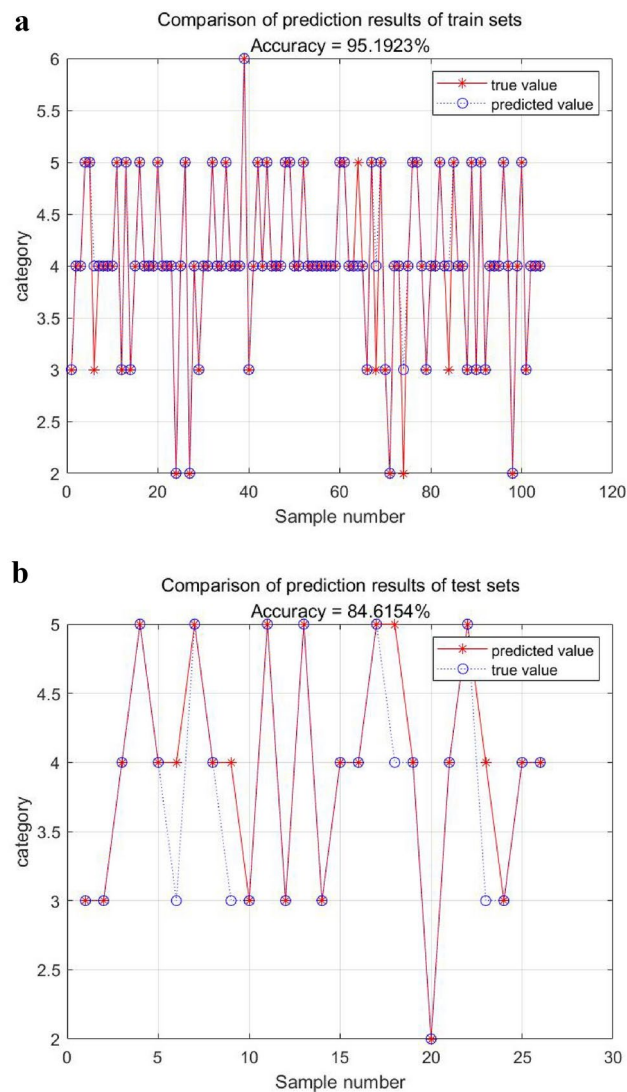


Fig. 9. (a) Comparison of the 4-point results of the classification training set based on the gravitational fireworks support vector machine. (b) Comparison of the 4-point results of the classification test set based on the gravitational fireworks support vector machine.

Through in-depth simulation experiments, we have extracted several pivotal insights. As shown in Table 5, the performance of the sequential fusion classification method is significantly better than that of the nonsequential fusion classification. This improvement, which is primarily quantified using benchmarks such as the mean squared error, can be attributed to the innovative capacity of the sequential method to comprehend and harness the continuity of data sequences and the intrinsic interconnections between data points—qualities not effectively addressed by traditional models.

In juxtaposition with prevalent techniques such as deep neural networks (*DNN*s) and recurrent neural networks (*RNN*s), our method underscores its versatility. While *DNN*s and *RNN*s are lauded for their profound representational power, they often necessitate substantial computational resources and extensive training datasets. In contrast, our sequential fusion classification approach strikes a harmonious balance between performance and computational demands. Additionally, the adaptability of our method is particularly advantageous in scenarios where dataset size and diversity may be constrained, positioning it as a formidable contender in the realm of classification models. The results clearly demonstrate that our proposed *DGFO-SVM* method outperforms the baseline SVM and other comparison algorithms in terms of accuracy and F1 score. Notably, *DGFO-SVM* achieves an accuracy of 91%, which is at least 2% higher than that of the best-performing alternative (*RNN*) and significantly better than that of the traditional alternative *SVM*. These results validate the effectiveness of integrating the differential gravitational fireworks optimization with *SVM* for air quality classification.

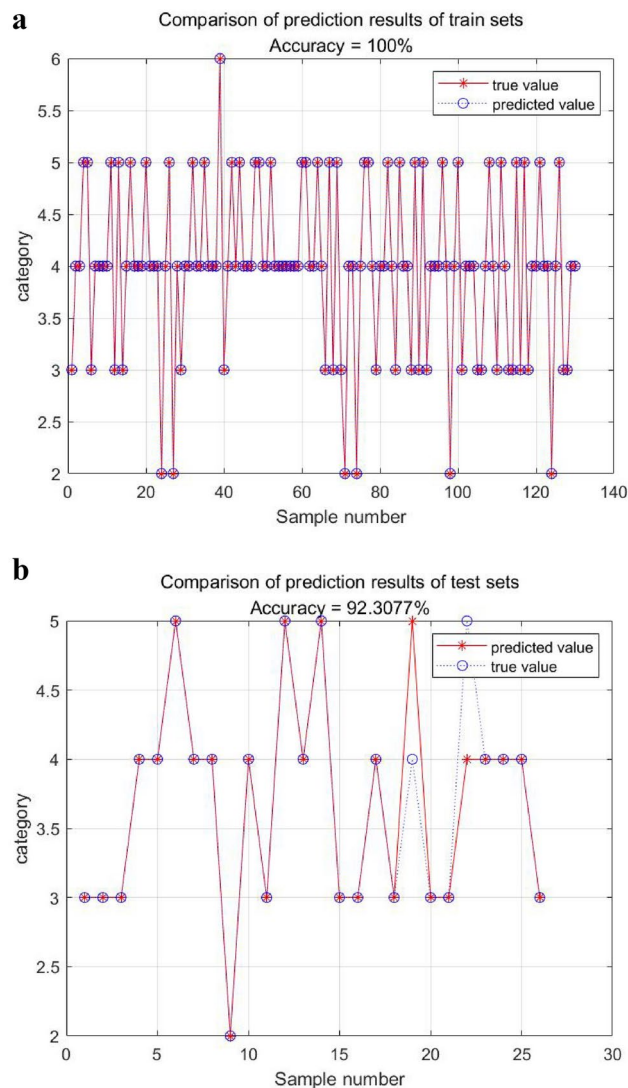


Fig. 10. (a) Comparison of the 5-point results of the classification training set based on the gravitational fireworks support vector machine. (b) Comparison of the 5-point results of the classification test set based on the gravitational fireworks support vector machine.

Results and analysis

In this study, we conduct scientific exploration and innovation, culminating in the development of an enhanced support vector machine (SVM) classification framework that is optimized through the integration of differential gravitational firework optimization. Our empirical analysis and simulation testing yield several significant contributions:

First, we integrate the second law of geography into the air quality analysis framework, offering a novel perspective that aids in uncovering and understanding the complex interactions between environmental and geographical factors. Compared with the use of nonordinal variables, the use of ordinal classification methods resulted in an accuracy improvement of 3 to 10% in air quality index classification.

Second, our framework demonstrated superior flexibility and precision in addressing nonlinear classification problems, confirming its exceptional ability to obtain optimal data mappings across multidimensional scales.

Additionally, the incorporation of differential gravitational firework optimization—especially in optimizing the penalty parameter c and the kernel function parameter g —significantly enhanced the prediction accuracy of pollution concentration classification, improving it by 3 to 15% and effectively mitigating premature convergence issues.

Through an in-depth analysis of multisource, redundant, and complementary air quality data, we quantitatively examined the spatiotemporal dimensions of the data, further increasing the classification accuracy. While we initially adopted a mean data interpolation strategy to address missing data, which has practical advantages, we recognize its limitations in more complex contexts. Therefore, we recommend that future research explore more advanced interpolation methods.

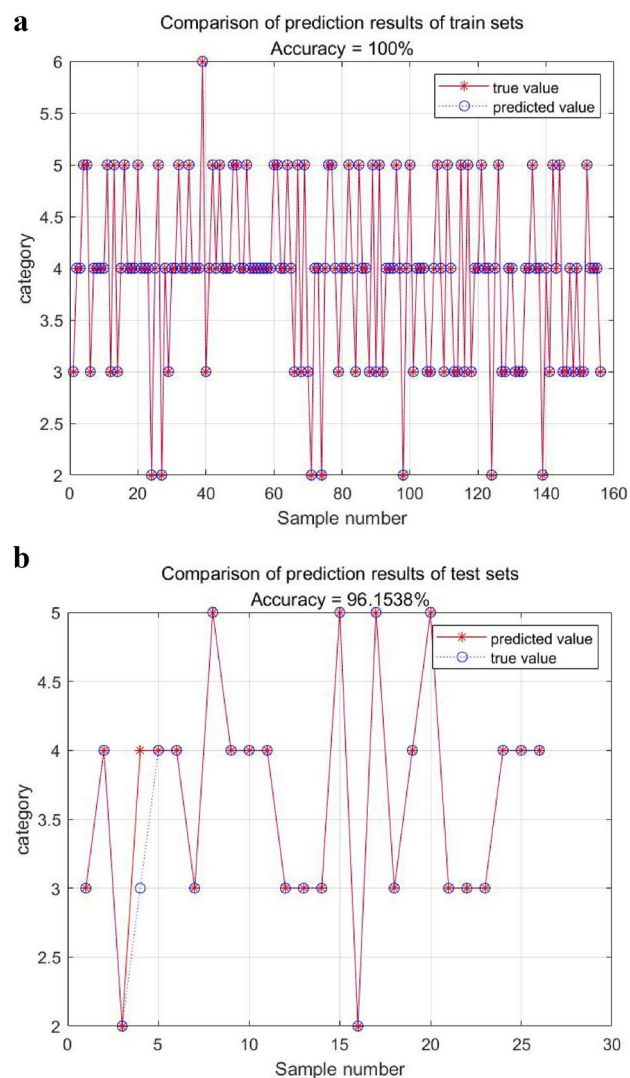


Fig. 11. (a) Comparison of the 6-point results of the classification training set based on the gravitational fireworks support vector machine. (b) Comparison of the 6-point results of the classification test set based on the gravitational fireworks support vector machine.

In conclusion, our study presents a new air quality monitoring method that demonstrates high efficiency and accuracy, with potential academic value and practical applications.

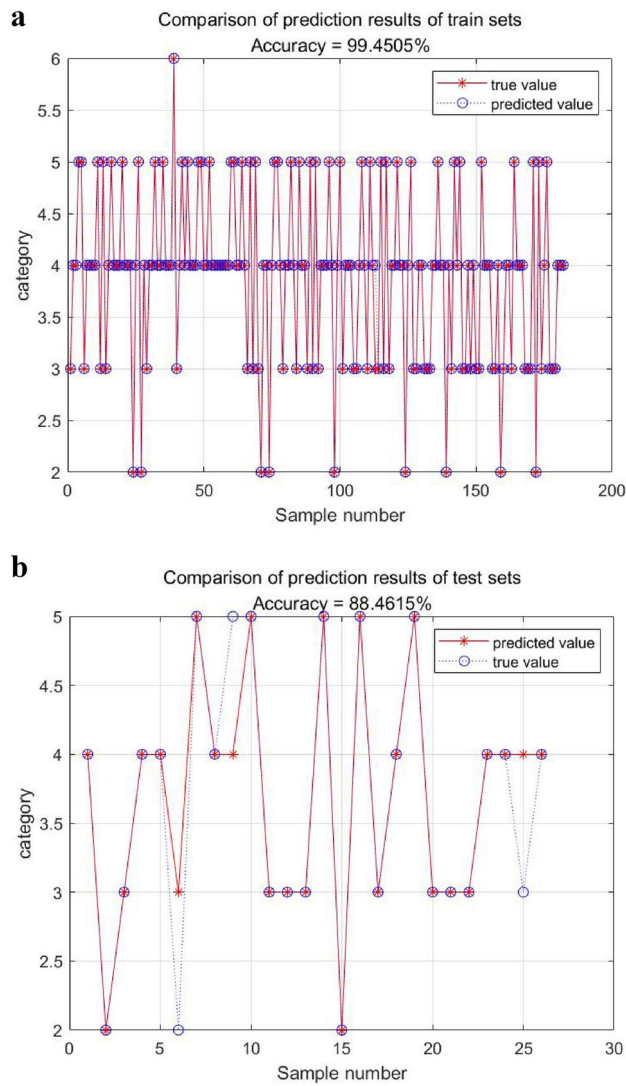


Fig. 12. (a) Comparison of the 7-point results of the classification training set based on the gravitational fireworks support vector machine. (b) Comparison of the 7-point results of the classification test set based on the gravitational fireworks support vector machine.

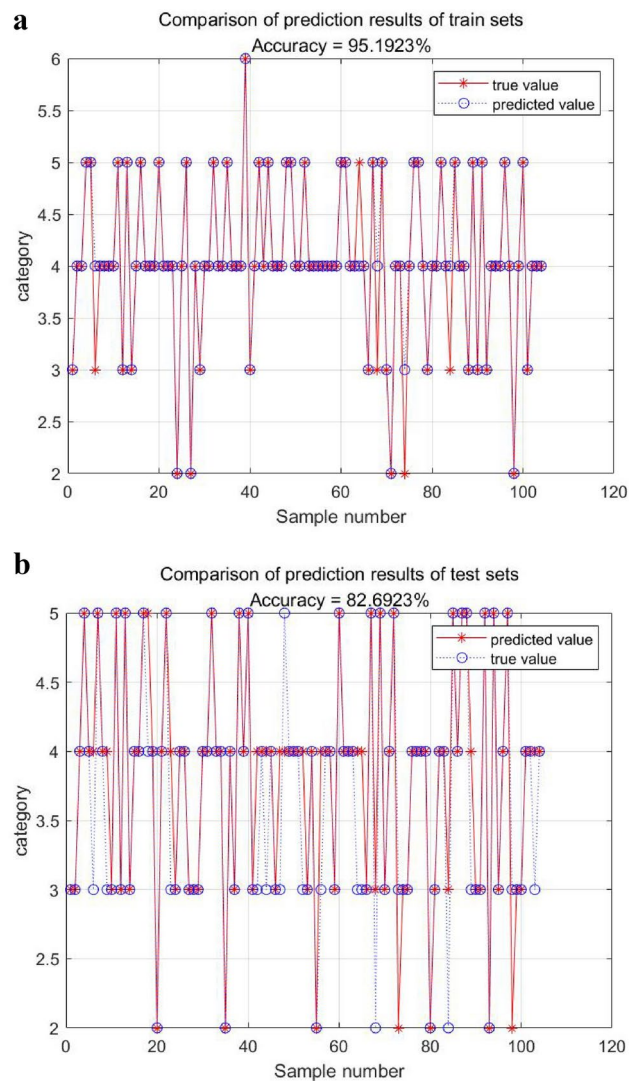


Fig. 13. (a) Comparison of the 4–7-point nonsequential classification training set results based on the gravitational fireworks support vector machine. (b) Comparison of the 4–7-point results of the nonsequential classification test set based on the gravitational fireworks support vector machine.

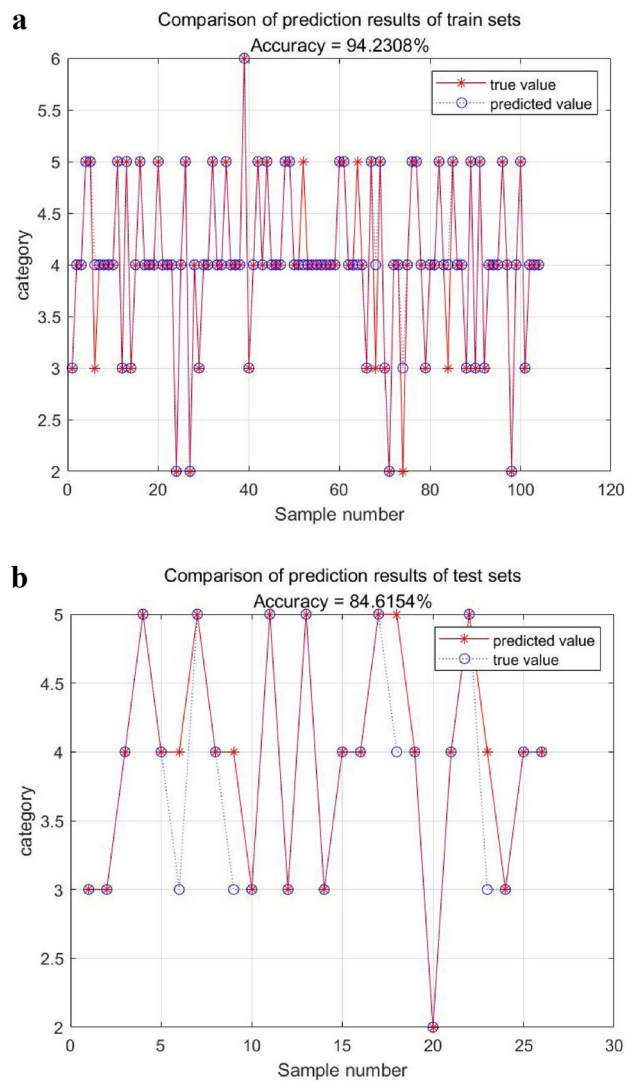


Fig. 14. (a) Comparison of the 4-point results of the classification training set based on the differential gravitational fireworks support vector machine. (b) Comparison of the 4-point results of the classification test set based on the differential gravitational fireworks support vector machine.

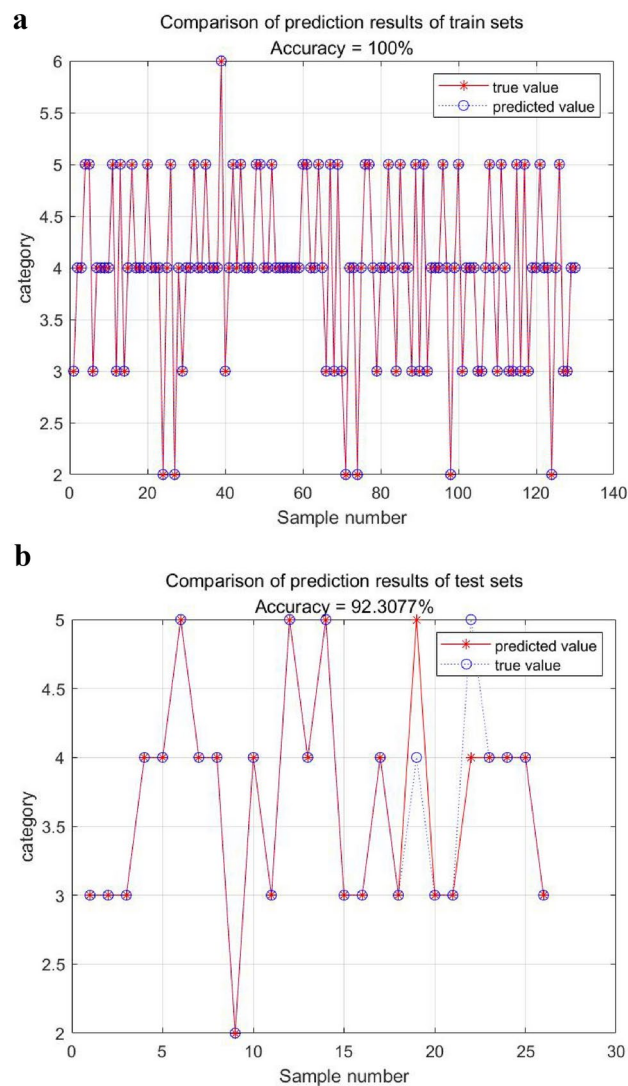


Fig. 15. (a) Comparison of the 5-point results of the classification training set based on the differential gravitational fireworks support vector machine. (b) Comparison of the 5-point results of the classification test set based on the differential gravitational fireworks support vector machine.

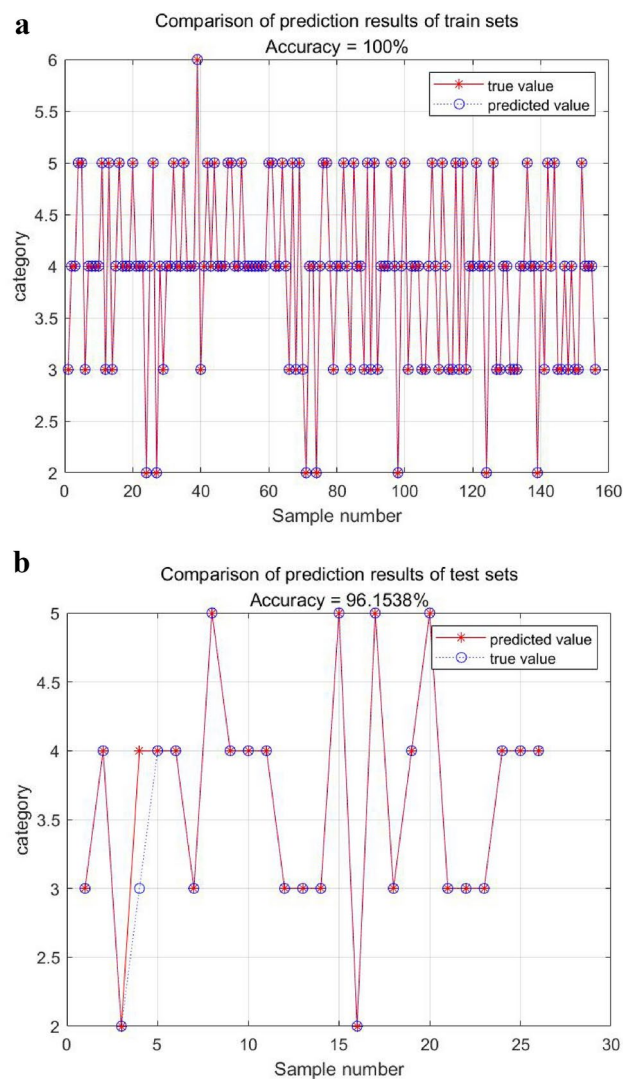


Fig. 16. (a) Comparison of the 6-point results of the classification training set based on the differential gravitational fireworks support vector machine. (b) Comparison of the 6-point results of the classification test set based on the differential gravitational fireworks support vector machine.

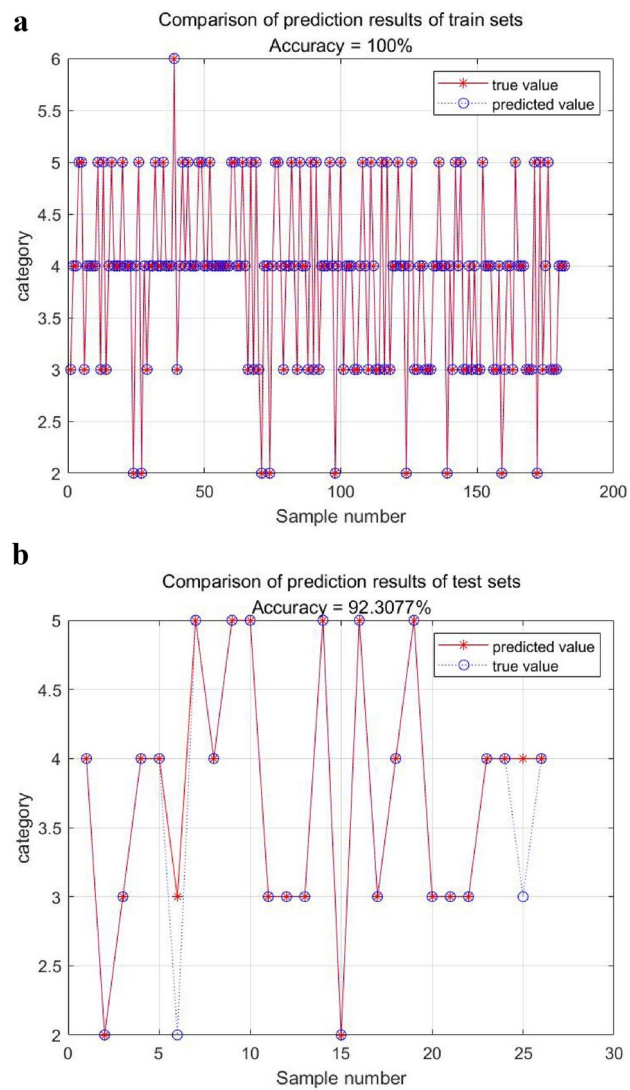


Fig. 17. (a) Comparison of the 7-point results of the classification training set based on the differential gravitational fireworks support vector machine. (b) Comparison of the 7-point results of the classification test set based on the differential gravitational fireworks support vector machine.

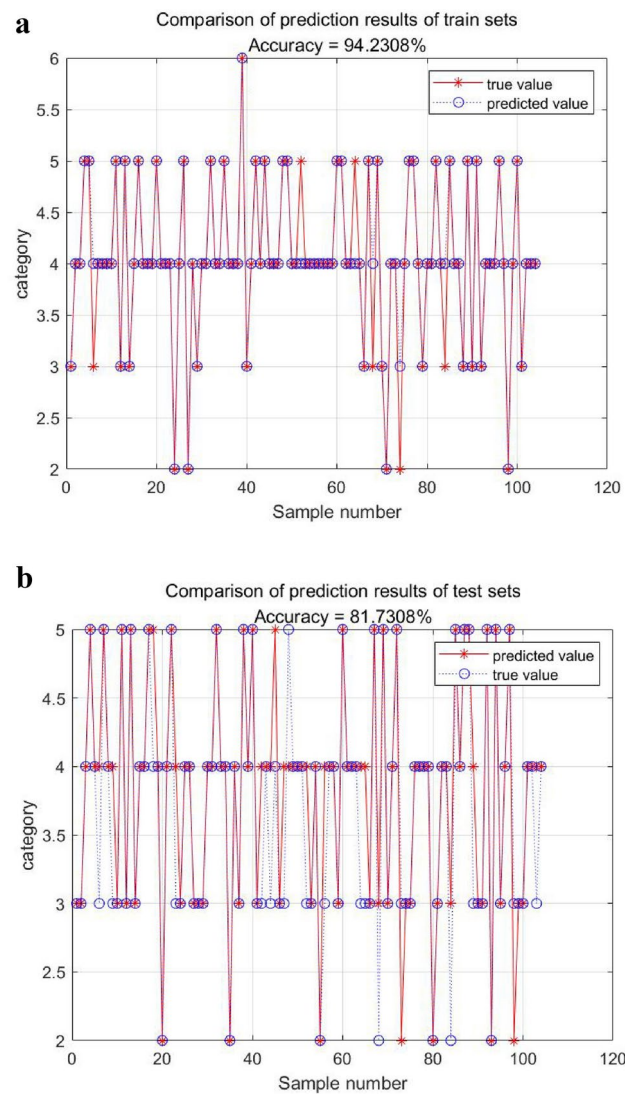


Fig. 18. (a) Comparison of the 4–7-point results in the nonsequential classification training set based on the differential gravitational fireworks support vector machine. (b) Comparison of the 4–7-point results in the nonsequential classification test set based on the differential gravitational fireworks support vector machine.

Accuracy of sequential classification	SVM	Gravitational fireworks SVM	Differential gravitational fireworks SVM
0–3→4 train	89.4231	95.1923	94.2308
0–4→5 train	90	100	100
0–5→6 train	89.7436	98.7179	100
0–6→7 train	88.4615	99.4505	100
Average accuracy rate	89.40705	98.340175	98.5577
0–3→4–7 train	89.4231	95.1923	94.2308
0–3→4 test	84.6154	84.6154	84.6154
0–4→5 test	73.0769	92.3077	92.3077
0–5→6 test	76.9231	84.6154	96.1538
0–6→7 test	73.0769	92.3077	92.3077
Average accuracy rate	76.923075	88.46155	91.34615
0–3→4–7 test	73.0769	82.6923	81.7308

Table 5. Comparison of the accuracy results of the fusion classification methods.

Algorithm	Accuracy (%)
SVM	82.1
GFO-SVM	85.3
DGFO-SVM (proposed method)	91.0
Deep neural network (DNN)	88.5
Recurrent neural network (RNN)	89.2
XGBoost	87.4
Random forest (RF)	84.7

Table 6. Comparison of classification accuracy.

Algorithm	F1 Score
SVM	0.79
GFO-SVM	0.82
DGFO-SVM (proposed method)	0.89
Deep neural network (DNN)	0.85
Recurrent neural network (RNN)	0.87
XGBoost	0.84
Random forest (RF)	0.81

Table 7. Comparison of F1 scores.

Data availability

The data supporting this study are available through the Qingyue Open Environment Data Centre website. Notably, the availability of these data is limited, and they have been authorized for use only in the context of this study; hence, they are not publicly accessible. However, interested individuals or organizations can request access to these data by paying a reasonable fee. For data acquisition, please visit “https://data.epmap.org/product/province_air”. The website provides detailed information on how to apply for the data, the associated costs, and contact information. The guidelines provided on the website should be followed to facilitate prompt processing of a request. The pollution concentration standards and air quality index grading standards for different substances come from relevant regulations set by the National Environmental Protection Administration of China.

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Author contributions

All authors contributed to the study conception and design. Chao Bian wrote the data analysis and the Introduction section. Writing—Guangqiu Huang finished reviewing and editing. All authors read and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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